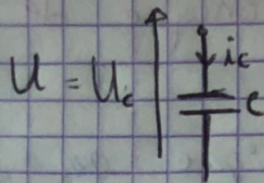


Série 1

Exo 1:

①



$$i_c = C \frac{dU_c}{dt}$$

$$\int f f' = \frac{1}{2} f^2$$

$$U = U_{\max} \sin \omega t \Rightarrow \text{périodique}$$

$$\langle s \rangle = \frac{1}{T} \int_0^T s(t) dt$$

$$P_{\text{moy}} = \langle p \rangle = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T u(t) i(t) dt$$

$$P_{\text{moy}} = \langle p \rangle = \frac{1}{T} \int_0^T U_{\max} \sin \omega t \times C U_{\max} \omega \cos \omega t dt$$

$$= \frac{U_{\max}^2 \cdot C \cdot \omega}{T} \int_0^T \sin \omega t \cos \omega t dt$$

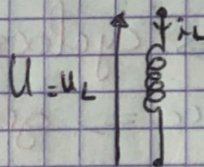
$$= \frac{U_{\max}^2 \cdot C \cdot \omega}{T} \left[\frac{1}{2\omega} \sin^2 \omega t \right]_0^T$$

$$= \frac{C \cdot \omega \cdot U_{\max}^2}{2\omega T} \left[\sin^2 \omega T - \sin^2 \omega 0 \right]$$

$$\sin^2 \frac{2\pi \cdot T}{T} = 0$$

donc $\{ P_{\text{moy}} = \langle p \rangle = 0 \}$

②



$$U = U_{\max} \cos \omega t = U_{\max} \sin \omega t \Rightarrow \text{périodique}$$

$$\langle s \rangle = \frac{1}{T} \int_0^T s(t) dt$$

$$P_{\text{moy}} = \langle p \rangle = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T u(t) i(t) dt$$

$$U_L = L \frac{di_L}{dt} \Rightarrow i_L = \frac{1}{L} \int u_L(t) dt$$

$$P_{\text{moy}} = \frac{1}{T} \int_0^T U_{\max} \sin \omega t \times -\frac{U_{\max}}{L} \cos \omega t dt$$

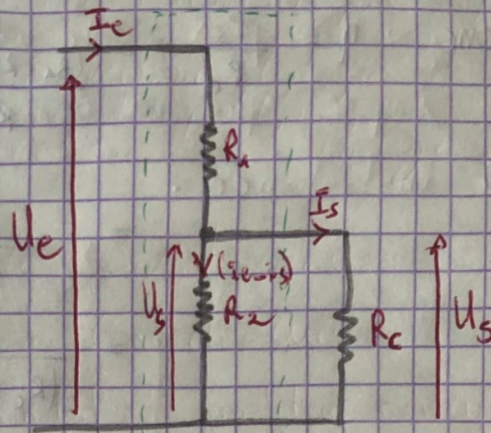
$$= -\frac{U_{\max}^2}{T L \omega} \int_0^T \sin \omega t \cos \omega t dt$$

$$= -\frac{U_{\max}^2}{T L \omega} \left[\frac{1}{2\omega} \sin^2 \omega t \right]_0^T$$

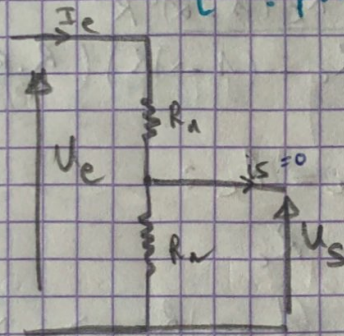
$$P_{\text{moy}} = \langle p \rangle = \frac{-U_{\text{moy}}}{T\omega} [\sin^2 \omega T - \sin^2 \omega 0]$$

donc $\{ P_{\text{moy}} = \langle p \rangle = 0 \}$

Exo 2



① l'expression de rendement η :
 $\eta = ?$ pour $R_1 = R_2$, $R_c = \infty$, sortie ouverte



$$\text{on } \eta = \frac{P_s}{P_e} = \frac{U_s \cdot i_s}{U_e \cdot i_e}$$

$$= \frac{R_2 \cdot i_e}{(R_1 + R_2) \cdot i_e} = \frac{R_2}{R_1 + R_2}$$

$$\eta = \frac{1}{2} = 0,5 = 50\%$$

② l'expression de rendement η en fct de R_1 , R_2 et R_c

$$\eta = \frac{P_s}{P_e} = \frac{U_s \cdot i_s}{U_e \cdot i_e}$$

$$i_s \times [U_s = R_c \cdot i_s = R_2 (i_e - i_s) = (R_c \parallel R_2) i_e]$$

$$i_e \times [U_e = R_2 (i_e - i_s) + R_1 \cdot i_e = (R_c \parallel R_2) i_e + R_1 \cdot i_e]$$

$$= R_c \cdot i_s + R_1 \cdot i_e$$

• on a $R_c \cdot i_s = R_2 (i_e - i_s) \Rightarrow (R_c + R_2) i_s = R_2 i_e$

$$i_s = \frac{R_2}{R_c + R_2} i_e$$

$$P_s = U_s \cdot i_s = R_c \cdot i_s \cdot i_s = R_c i_s^2$$

$$P_e = [R_1 i_e + R_2 (i_e - i_s)] i_e = (R_1 + R_2) i_e^2 - R_2 i_s i_e$$

$$P_e = (R_1 + R_2) \left(\frac{R_c + R_2}{R_2} \right)^2 i_s^2 - R_2 \left(\frac{R_c + R_2}{R_2} \right) i_s^2$$

$$\text{donc } \eta = \frac{P_s}{P_e} = \frac{R_c}{\frac{(R_1 + R_2)(R_c + R_2)^2}{R_2^2} - (R_c + R_2)}$$

$$= \frac{R_c \cdot R_2^2}{(R_1 + R_2)(R_c + R_2)^2 - (R_c + R_2) \cdot R_2^2}$$

$$= \frac{R_c \cdot R_2^2}{(R_c + R_2)[(R_1 + R_2)(R_c + R_2) - R_2^2]}$$

$$= \frac{R_c \cdot R_2^2}{(R_c + R_2)(\underbrace{(R_1 + R_2) R_c}_R + R_1 R_2)}$$

$$\eta = \frac{R_c (1 - \alpha)^2 R^2}{(R_c + (1 - \alpha)R)(R \cdot R_c + \alpha(1 - \alpha)R^2)}$$

③ on calcule η pour $R_1 = R_2 = R_c$:

$$R_1 = R_2 = R_c$$

$$\alpha R = (1 - \alpha)R \Rightarrow \alpha = \frac{1}{2}$$

$$R_c = \frac{1}{2} R$$

$$\eta = \frac{1}{6} = 0,16 = 16\%$$

Série 01

Exo 3 :

①

on a $V_T + R_L I_T = +V_{CC} \Rightarrow I_T = \frac{V_{CC} - V_T}{R_L} = \frac{23}{27 \cdot 10^3}$
 $\Rightarrow$ le thyristor ne peut être amorcé, il est bloqué
 Car : $\{ I_T = 0,8 \text{ mA} < I_H = 1 \text{ mA} \}$
 = 0,8 mA $\leq$ I_H
 le courant de maintien

②

pour que le thyristor sera amorcé il faut que $I_T > I_H$

on a $R_L = \frac{V_{CC} - V_T}{I_T}$
 $\Rightarrow R_{L \max} = \frac{V_{CC} - V_T}{I_{T \min}} = \frac{23}{I_H} = 23 \text{ K}\Omega$
 doctère minimal

Exo 4 :

①

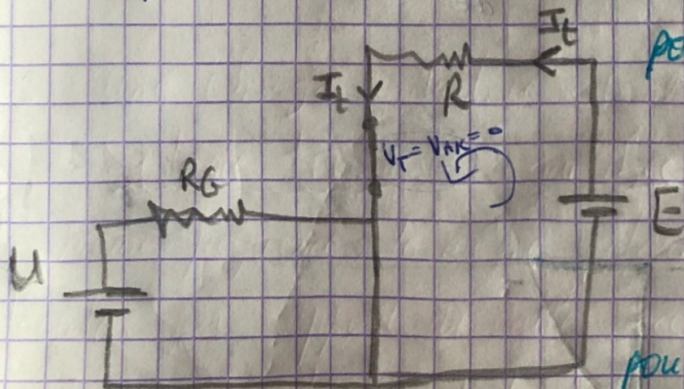
on a $V_{GK} = 0,8 \text{ V}$, $I_{GT} = 0,1 \text{ mA}$
 on a $U = R_G \cdot I_G + V_{GK}$

$$U = 0,8 + 5 \cdot 10^3 \times 0,1 \cdot 10^{-3} = 5,8 \text{ V}$$

A partir de la tension 5,8 V, la tension de gâchette dépasse la tension minimale donc le thyristor devient passant.

②

$V_T = 0 = V_{AK}$ donc le circuit devient:



pour $I_H = 10 \text{ mA}$

$$E = R \cdot I_T$$

$$\Rightarrow R = \frac{E}{I_T = I_H}$$

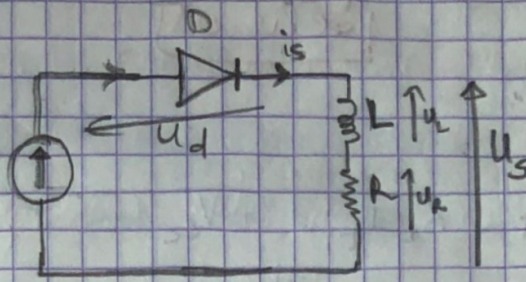
$$= \frac{24}{10 \cdot 10^{-3}} = 2,4 \text{ K}\Omega$$

pour $I_H = 500 \text{ nA}$

$$R = \frac{24}{500 \cdot 10^{-9}} = 48 \text{ K}\Omega$$

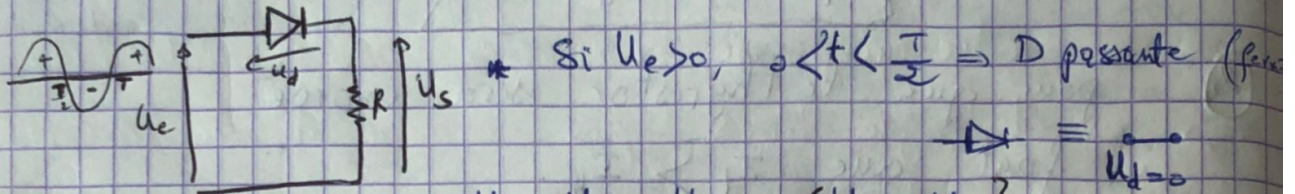
(Série 2)

Exo 1:



Charge inductive

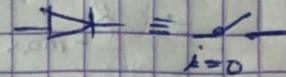
$$u_e = U_{max} \sin \omega t = U_{eff} \sqrt{2} \sin \omega t$$



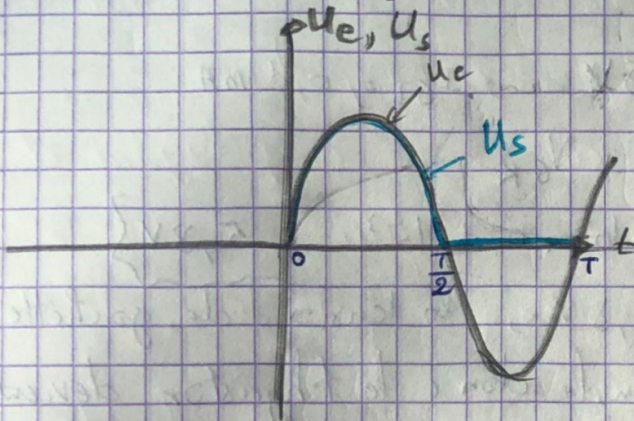
$$u_s + u_d = u_e \Rightarrow \{u_s = u_e\}$$

* Si $u_e < 0$, $\frac{T}{2} < t < T \Rightarrow D$ bloqué (ouvert)

$$u_s = R \cdot i_s = 0$$

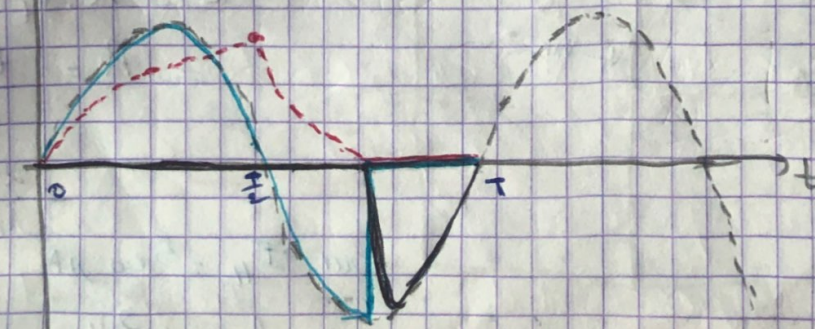


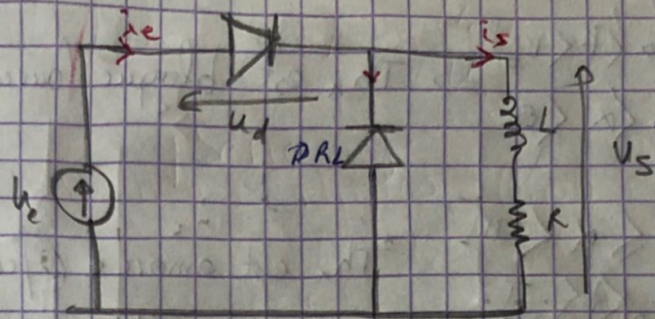
$$u_s + u_d = u_e \Rightarrow \{u_d = u_e\}$$



$$\begin{cases} u_s + u_d = u_e \\ u_s = u_L + u_R = L \frac{di_s}{dt} + R \cdot i_s \end{cases}$$

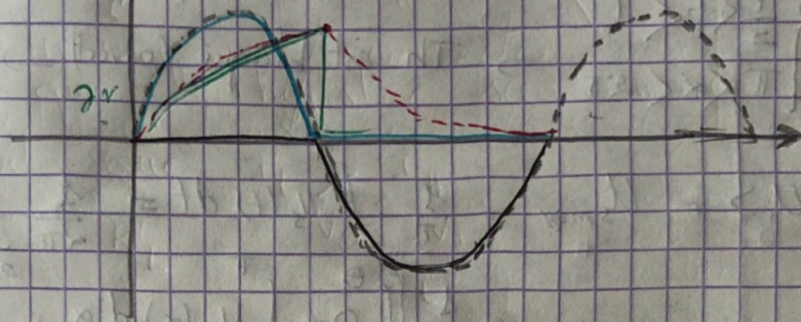
u_e, i_s, u_s, u_d





$u_e > 0 \Rightarrow D \text{ bloque} \Rightarrow DRL \equiv \text{---}$

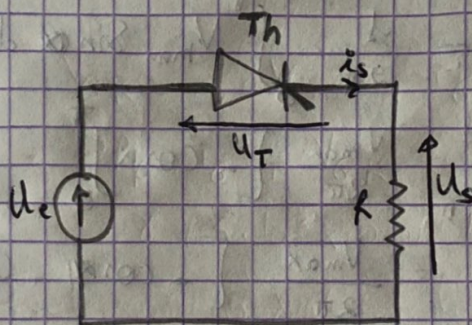
$u_e < 0 \Rightarrow D \text{ passe} \Rightarrow DRL \equiv \text{---}$
 $\rightarrow u_e, i_s, u_s, u_d$



* phase de redressement : courant tourne dans la même place.

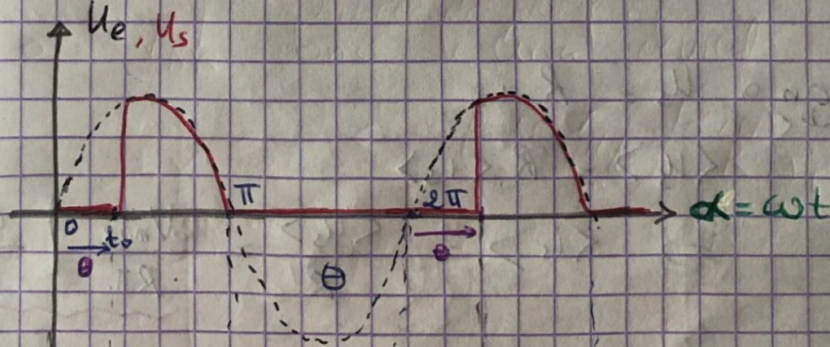
Exo 2)

①



$$u_e = U_{\max} \sin \omega t = U_{\text{eff}} \sqrt{2} \sin \omega t$$

$$u_e = U_{\max} \sin \theta$$



$$U_{\max} = \frac{U_{\text{eff}} \sqrt{2}}{R}$$

$$\bullet \begin{cases} 0 < t < t_0 \\ 0 < \alpha < \theta \end{cases} \Rightarrow \text{Th est bloqué (ouvert)} \rightarrow i_s = 0 \Rightarrow \underline{U_s = R i_s = 0}$$

$$\bullet \begin{cases} \theta \leq \alpha < \pi \\ t_0 < t < \frac{T}{2} \end{cases} \Rightarrow \text{Th est amorcé (fermé)} \Rightarrow U_T = 0$$

$$\{ U_s = U_e \}$$

$$U_s = R \cdot i_s = U_e$$

$$\{ i_s = \frac{U_e}{R} \}$$

$$\bullet \begin{cases} \frac{T}{2} < t < T \\ \pi < \alpha < 2\pi \end{cases} \Rightarrow \text{Th est bloqué (ouvert)} \Rightarrow i_s = 0 \Rightarrow \underline{U_s = R i_s = 0}$$

②

$$\langle U_s \rangle = U_{s \text{ moy}} = \frac{1}{T} \int_0^T U_s(t) dt = \frac{1}{2\pi} \int_0^{2\pi} U_s(\alpha) d\alpha$$

$$= \frac{1}{2\pi} \left[\int_0^\theta 0 d\alpha + \int_\theta^\pi U_e(\alpha) d\alpha + \int_\pi^{2\pi} 0 d\alpha \right]$$

$$U_{s \text{ moy}} = \frac{1}{2\pi} \int_\theta^\pi U_e(\alpha) d\alpha$$

$$= \frac{1}{2\pi} \int_\theta^\pi V_{\text{max}} \sin \alpha d\alpha$$

$$= \frac{V_{\text{max}}}{2\pi} \left[-\cos \alpha \right]_\theta^\pi$$

$$= \frac{V_{\text{max}}}{2\pi} \left[-\cos \pi + \cos \theta \right]$$

$$\{ U_{s \text{ moy}} = \frac{V_{\text{max}}}{\pi} \left[\frac{\cos \theta + 1}{2} \right] \}$$

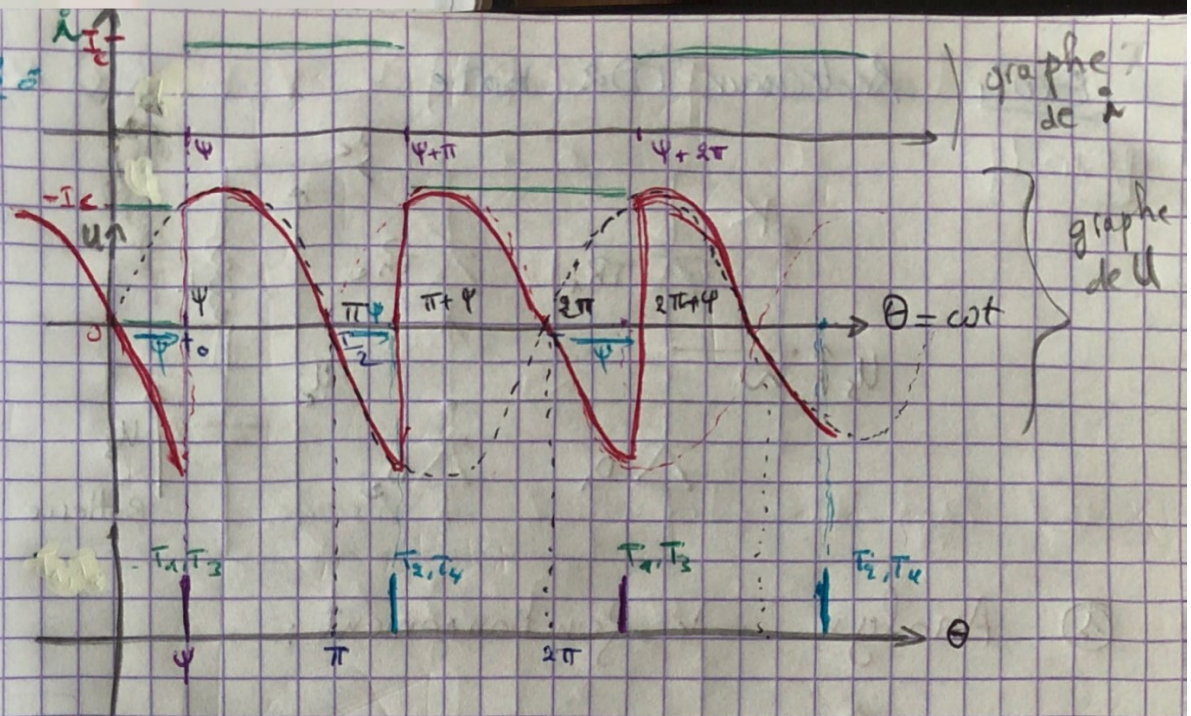
$$\bullet \text{ On a } U_s = R \cdot i_s$$

$$\langle U_s \rangle = R \langle i_s \rangle$$

$$\Rightarrow \langle i_s \rangle = \frac{\langle U_s \rangle}{R} = \frac{V_{\text{max}}}{R\pi} \left[\frac{\cos \theta + 1}{2} \right]$$

EXO 3

a)



$\begin{cases} t_0 < K \frac{T}{2} \\ \psi < \theta < \pi \end{cases} \Rightarrow Th_1 \text{ et } Th_2 \text{ sont amorcées (on les remplace par un fils)} \text{ et } Th_3, Th_4 \text{ par } \text{---}$
 donc $u_c(t) = u(t)$

$$i = i_T = I_c = i_T$$

$\psi + \pi < \theta < 2\pi \Rightarrow Th_2, Th_4 \text{ sont amorcés (fermés)}$

$Th_3, Th_1 \text{ s'ouvrent}$

$$\text{donc } u_c(t) + u(t) = 0$$

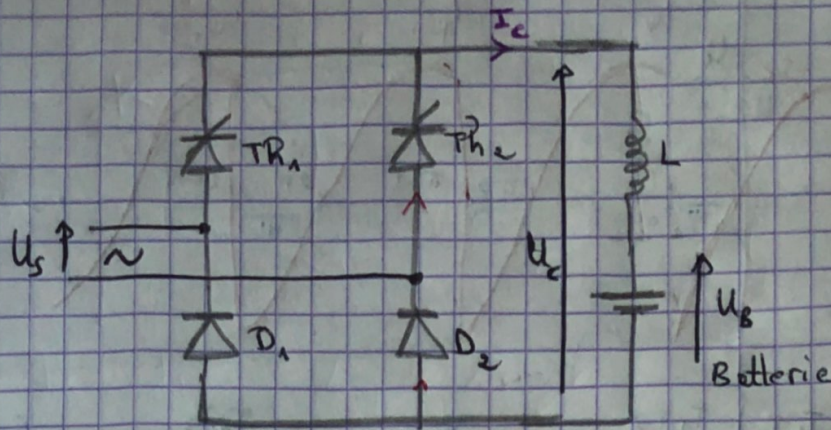
$$\Rightarrow u_c(t) = -u(t)$$

$$i = -I_c$$

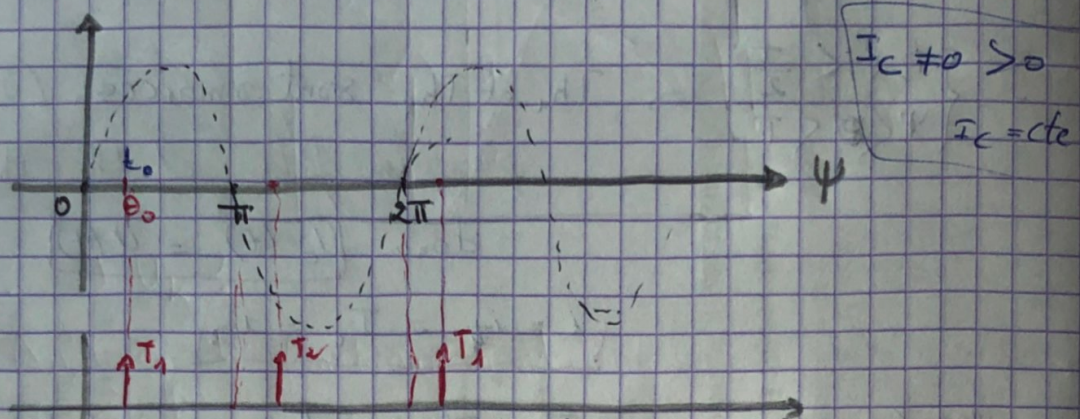
②

$$u_{cmoy} = \frac{1}{\pi} \int_{\psi}^{\psi + \pi} u_c(\theta) d\theta$$

Exo 1: Redresseur PD2 à 2V



① Analyse de fonctionnement:



- $\psi < \theta_0$
- $\theta_0 < \psi < \pi$: D_2 fermée, T_1 ouvert
 D_1 ouvert, T_2 fermée

$$\Rightarrow i_{D1} = i_{T1} = i_{T2} = 0 \Rightarrow \underline{I_c = 0}$$

impossible

dans ce cas T_2 reste fermée, T_1 ouvert
donc $\boxed{U_c = 0}$ c'est une phase de roue libre.
charge isolée.

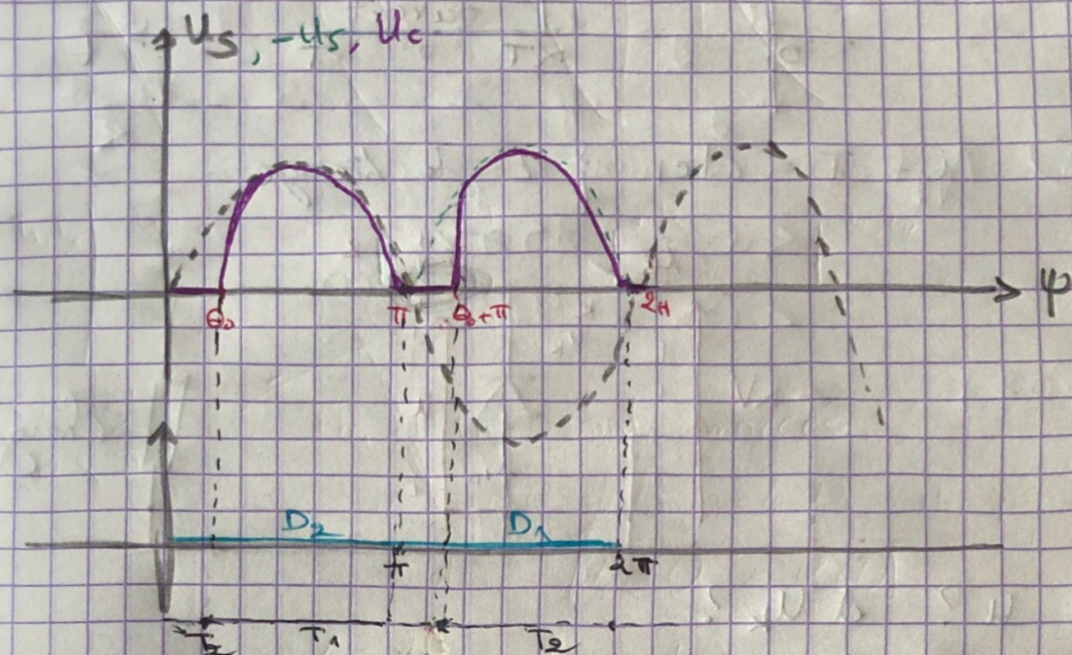
- $\theta_0 < \psi < \pi$: D_2 ferme, T_1 fermé
 $i = i_{T1} = I_c = i_{D2}$
 $\Rightarrow \boxed{U_c = U_s}$

- $\pi < \psi < \theta_0 + \pi \Rightarrow D_1$ fermé, T_2 ouvert $\Rightarrow T_1$ reste amorcée
 $\Rightarrow \boxed{U_c = 0}$ phase de roue libre

• $\pi + \theta_0 < \varphi < 2\pi \rightarrow T_2$ amorcée et D_1 fermée

$$\Rightarrow U_c = -U_s$$

$$I_c = i_{T_2} = i_{D_1} = -i$$



$[0, \theta_0] \rightarrow T_2$ et D_2

$[\theta_0, \pi] \rightarrow T_1$ et D_2

$[\pi, \pi + \theta_0] \rightarrow T_1$ et D_1

$[\pi + \theta_0, 2\pi] \rightarrow D_1$ et T_2

$$U_{cmoy} = \frac{1}{\pi} \int_{\theta_0}^{\pi} U\sqrt{2} \sin \varphi d\varphi$$

$$\text{on a } \omega t = \varphi$$

$$dt = \frac{1}{\omega} d\varphi$$

$$t = t_0 \rightarrow \varphi = \theta_0$$

$$t = \frac{\pi}{\omega} \rightarrow \varphi = \pi$$

$$U_{cmoy} = \frac{1}{\pi} \int_{\theta_0}^{\pi} U\sqrt{2} \sin \varphi d\varphi$$

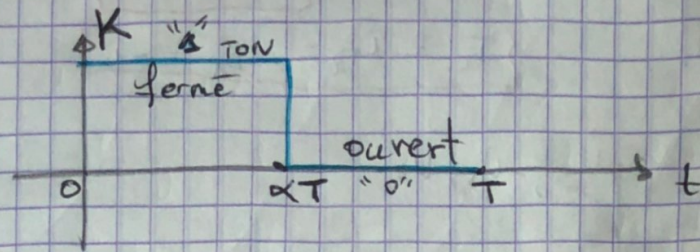
$$= \frac{U\sqrt{2}}{\pi} \int_{\theta_0}^{\pi} \sin \varphi d\varphi$$

$$= \frac{U\sqrt{2}}{\pi} [-\cos \varphi]_{\theta_0}^{\pi} = \frac{U\sqrt{2}}{\pi} [-\cos \pi + \cos \theta_0]$$

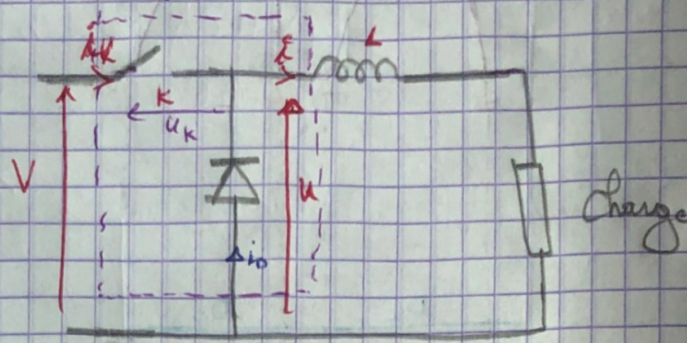
$$= \frac{U\sqrt{2}}{\pi} (1 + \cos \theta_0)$$

Série 3

Exo 1



$i = T = cte$ { L de lissage



③ u ? et u_K ?

Analyse de fonctionnement.

- Phase 1 : K fermé de $0 < t < \alpha T$:
Diode ouverte $i_D = 0$

$$u_K = 0$$

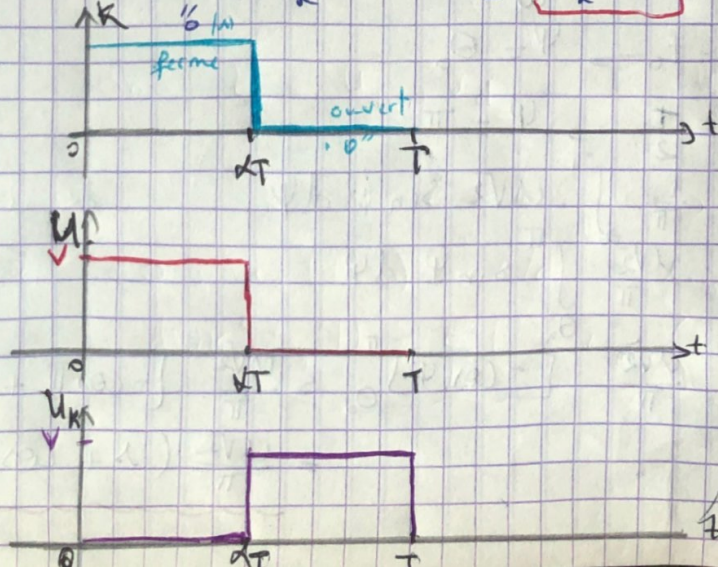
$$\Rightarrow i_K = i \quad , \quad u + u_K = V \Rightarrow \boxed{u = V}$$

- Phase 2 : K ouvert de $\alpha T < t < T$

$$i_K = 0 \Rightarrow i = i_D \Rightarrow D \text{ fermée (passant)}$$

$$\text{donc } \boxed{u = 0}$$

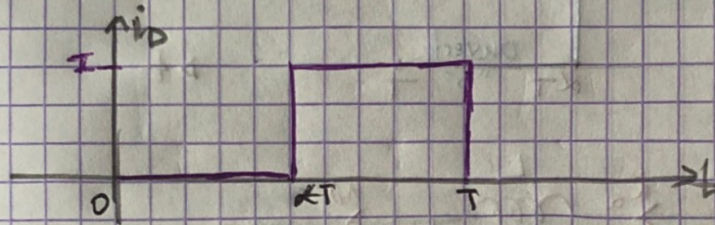
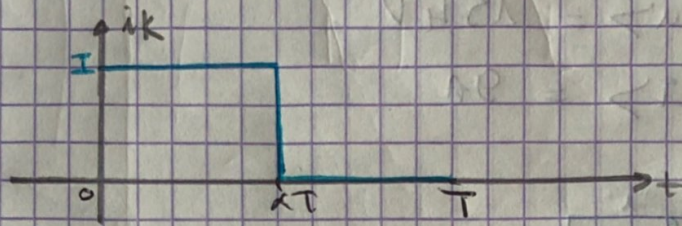
$$u + u_K = V \Rightarrow \boxed{u_K = V}$$



③ la valeur moyenne de u en fct de V et α

$$\begin{aligned}
 \langle u_{\text{moy}} \rangle &= \langle u \rangle = \frac{1}{T} \int_0^T u(t) dt \\
 &= \frac{1}{T} \left[\int_0^{\alpha T} u(t) dt + \int_{\alpha T}^T 0 dt \right] \\
 &= \frac{1}{T} \int_0^{\alpha T} V dt \\
 &= \frac{1}{T} \left[Vt \right]_0^{\alpha T} = \frac{\alpha VT}{T} = \alpha V
 \end{aligned}$$

④



④ les valeurs moyennes de i_k et i_D en fct de I et α

$$\begin{aligned}
 i_{k\text{moy}} &= \langle i_k \rangle = \frac{1}{T} \int_0^T i_k(t) dt \\
 &= \frac{1}{T} \left[\int_0^{\alpha T} I dt + \int_{\alpha T}^T 0 dt \right] \\
 &= \frac{1}{T} \left[It \right]_0^{\alpha T} \\
 \langle i_k \rangle &= \frac{I \alpha T}{T} = I \alpha
 \end{aligned}$$

$$\begin{aligned}
 i_{D\text{moy}} &= \langle i_D \rangle = \frac{1}{T} \int_0^T i_D(t) dt \\
 &= \frac{1}{T} \left[\int_0^{\alpha T} 0 dt + \int_{\alpha T}^T I dt \right] \\
 &= \frac{1}{T} \left[It \right]_{\alpha T}^T \\
 &= \frac{1}{T} [IT - I\alpha T] \\
 \langle i_D \rangle &= I(1 - \alpha)
 \end{aligned}$$

⑤ l'intensité I en fct de V, E, R et α

$$u = E + R \cdot i + L \frac{di}{dt}$$

$$u = E + R \cdot i \Rightarrow \langle u \rangle = E + R \langle i \rangle$$

$$u_{\text{moy}} = E + R i_{\text{moy}} = E + R I$$

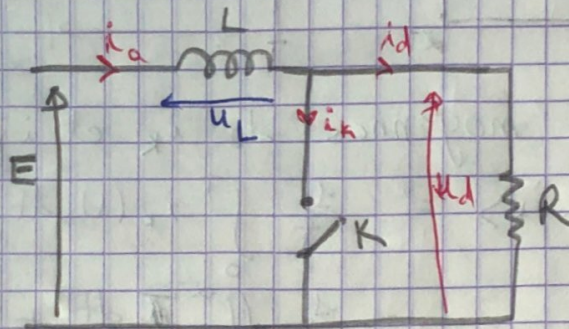
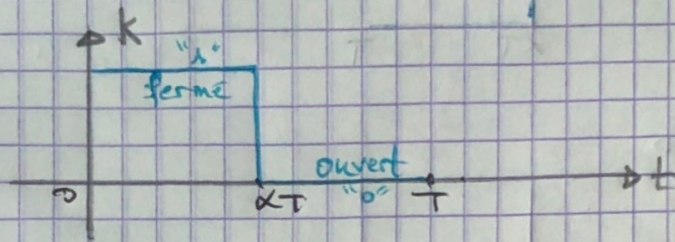
$$i_{\text{moy}} = I_{\text{eff}} = I \Rightarrow \boxed{I = \frac{V\alpha - E}{R} = 9A}$$

⑥ Applications numériques

$$\langle u \rangle = 154V$$

$$\langle I \rangle = 9A$$

Exo 2/



① Analyse de fonctionnement :

• phase 1 : K fermé : $0 < t < \alpha T$

$$\Rightarrow \boxed{u_k = 0 = u_d} \Rightarrow \boxed{i_d = 0}$$

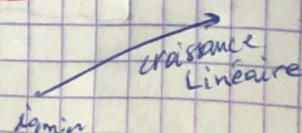
$$\Rightarrow \boxed{i_a = i_k}$$

$$\text{On a } E = u_L + u_d \Rightarrow E = u_L = L \frac{di_a}{dt}$$

$$\text{donc } E = L \frac{di_a}{dt} \xrightarrow{\text{intègre}} i_a(t) = \frac{E}{L} t + c$$

$$\text{à } t=0 : i_a(t=0) = c = i_{\text{amin}}$$

$$\text{donc } \boxed{i_a(t) = \frac{E}{L} t + i_{\text{amin}}}$$

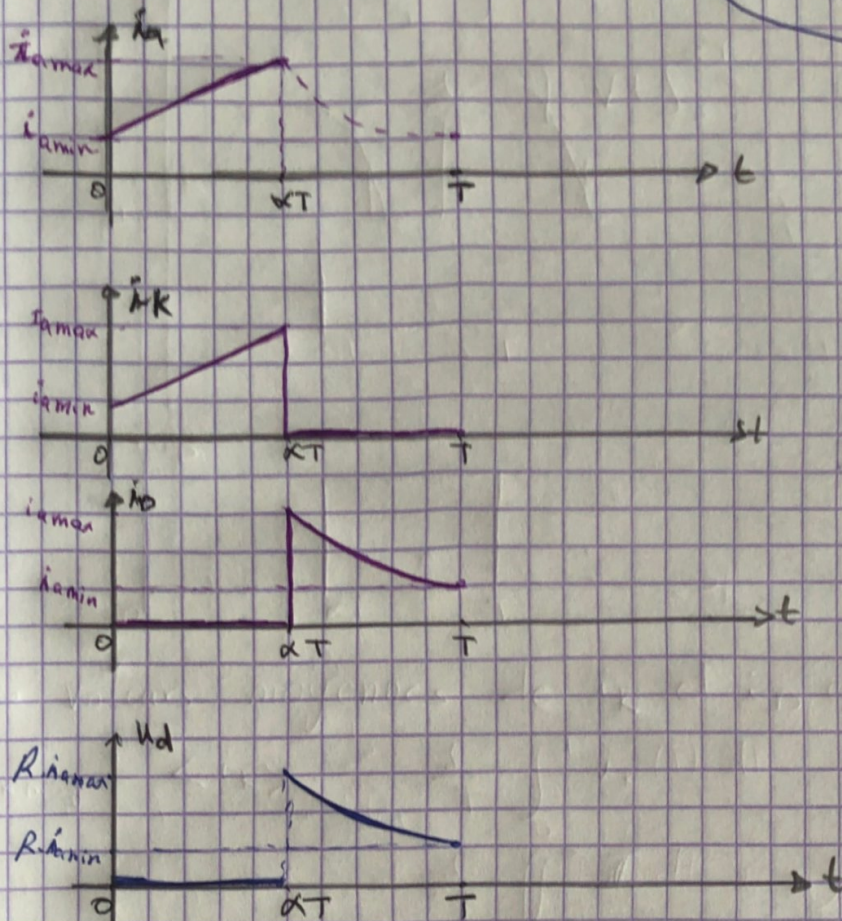


- phase 2: R ouvert $\alpha T < t < T$

$$\{i_k = 0\} \text{ et } \{i_a = i_d\} \quad \{u_d = R \cdot i_d\}$$

$$\Rightarrow E = U_d + U_L \Rightarrow E = R \cdot i_d + L \frac{di_d}{dt}$$

on a une décroissance exponentielle.



②

- Continuité du courant:

$$i_{a1}(t = \alpha T) = i_{a2}(t = \alpha T) \quad \text{Eq ①}$$

- Périodicité du courant:

$$i_{a1}(t = 0) = i_{a2}(t = T) \quad \text{Eq ②} \quad \begin{cases} i_{amin} = \\ i_{amax} = \end{cases}$$

$$* \langle i_d \rangle = i_{d\text{moy}} = \frac{1}{T} \int_0^T i_d(t) dt = \frac{1}{T} \int_{\alpha T}^T i_a(t) dt$$

$$\{ i_{d\text{moy}} = I_a \frac{1 - \alpha}{1} \}$$

$$* \langle u_d \rangle = R \cdot \langle i_d \rangle = \underline{R \cdot I_a (1 - \alpha)}$$

③